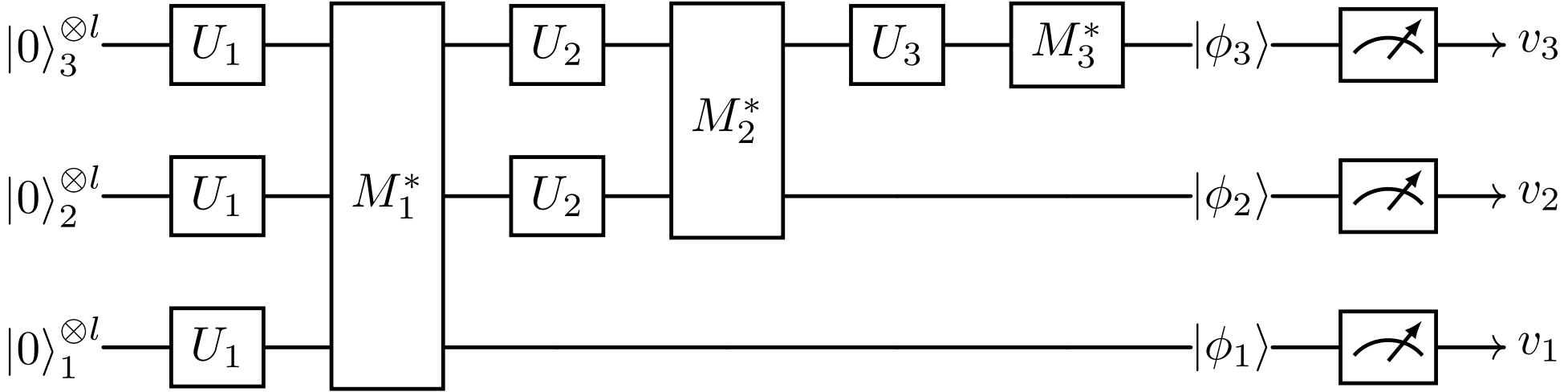
$|0\rangle_3^{\otimes l}$
$|0\rangle_2^{\otimes l}$
$|0\rangle_1^{\otimes l}$
U_1
U_1
U_1
M_1^*
U_2
U_2
M_2^*
U_3
M_3^*
$|\phi_3\rangle$
$|\phi_2\rangle$
$|\phi_1\rangle$
v_3
v_2
v_1